

Blockhouse Test Trial

Isabelle Haggquist | ichaggquist@gmail.com | +(720) – 204 – 0362

1 What’s the motivation behind measuring OFI at multiple depth levels of the order book?

The motivation behind measuring OFI at multiple depth levels of the order book is to capture a more comprehensive picture of the order book dynamics. The author’s integrated approach of combining OFIs from the top levels of the Limit Order Book into an integrated OFI variable provided a better explanation of price impact compared to using solely the best-level OFI. Furthermore, the study shows that once the information from these multiple levels is integrated into OFI, the inclusion of cross-asset OFI terms does not significantly improve the explanation of price impact. The strong correlations observed between different levels of OFI, with the first principal component explaining a large portion of the variance, also support the value of an integrated measure.

2 Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors used Lasso regression instead of OLS to estimate the primary cross-impact due to the characteristics of their data. With a large number of potential independent variables, OLS is deemed a poor fit for the data. Additionally, the presence of multicollinearity among cross-asset OFIs makes this data set unsuitable for OLS. Given Lasso’s ability to perform feature selection and regularization on high-dimensional and multicollinearity data, it is a good choice for this data set. Lasso is particularly useful for handling sparse data by shrinking their coefficients to zero. This improves model interpretability and predictability accuracy.

3 Why is OFI considered a better predictor of short-term returns than trade volume?

Seen in literature, Order Flow Imbalance is considered a significant driving force in price changes over short time intervals. The authors highlight previous studies demonstrating that price changes are mainly driven by OFI and that certain deep learning models trained on OFIs can significantly outperform those trained on order books or returns. Given these studies, the approach taken in this paper uses OFIs as the primary features in their analysis. Their findings further support OFI’s significance as a predictor of short term returns.